

Foundation (Jalapeno)

- Q1.** What is the slope of the line $y = 2x - 3$?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q2.** A line passes through points $(2, 3)$ and $(4, 5)$. What is its slope?
(A) 0.5
(B) 1
(C) 1.5
(D) 2
(E) 2.5
- Q3.** What does a slope of 0 represent?
(A) A vertical line
(B) A horizontal line
(C) A positive line
(D) A negative line
(E) A diagonal line
- Q4.** A line has a slope of -2 and passes through $(1, 3)$. What is its equation?
(A) $y = -2x + 5$
(B) $y = -2x + 1$
(C) $y = 2x - 5$
(D) $y = 2x + 1$
(E) $y = x - 2$
- Q5.** A graph shows a line with a slope of 3. If x increases by 2,
- what is the change in y ?
(A) 4
(B) 5
(C) 6
(D) 7
(E) 8
- Q6.** What is the slope of a line that passes through $(0, 0)$ and $(3, 4)$?
(A) $\frac{1}{3}$
(B) $\frac{4}{3}$
(C) 4
(D) 2
(E) 3
- Q7.** A line has an undefined slope. What can be said about this line?
(A) It is horizontal
(B) It is vertical
(C) It is diagonal
(D) It is curved
(E) It is a point
- Q8.** What is the slope of the line $x = 2$?
(A) 0
(B) 1
(C) 2
(D) undefined
(E) -1

Open Ended Questions (FRQ)

- Q9.** Find the equation of a line with a slope of 2 and passing through the point $(1, 3)$. Show your work and write the final equation in slope-intercept form $y = mx + b$. Use 5 lines or less.
- Q10.** A water tank is being filled at a rate of 2 gallons per minute. If the tank is currently empty, how much water will be in the tank after 5 minutes? Show your work and include units. Use 5 lines or less.

Chef's Tips

- Emphasize the importance of showing work for free response questions
- Remind students to use the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ when necessary
- Encourage students to check their work by plugging in points or using a graphing calculator